

Launch Vehicle Design optimization with Booster Recovery

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1 Abstract

The economics of launching satellites into space have been changed dramatically since SpaceX has introduced the highly successful Falcon 9 Launch Vehicle (LV) that is able to recover the most expensive component of the LV - the first stage booster. Following the success of the Falcon 9, many other LVs with similar goals are in different design and testing phases. Designing a LV includes the a complex compromise on a multi-disciplinary optimization. This paper presents an optimization process that takes the primary parameters that influence the performance of a LV and incorporate them all in a single optimization process to produce a one-stop preliminary design tool for reusable LVs.

2 Introduction

Launching a satellite into earth orbit is a very difficult engineering task that only a handful of countries have achieved. Like every aerospace product design, in general, LVs try to maximize the performance by optimizing the design, using small safety factors as much as possible. A LV is the result of a multi-disciplinary design that include propulsion, structural, aerodynamic, guidance and control (G&C), thermal, electrical (and many more) [1]. The nature of such design process is iterating over every element of the LV and slowly optimizing the LV as a whole [2]. The purpose of a LV is to put a given mass into a predefined orbit around earth or the sun. After every design iteration, begins a long performance analysis

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cycle of the proposed configuration. Usually, this step used to be a manual process where the engineers suggested a guidance law that tries to maximize performance, but this is limited by the flexibility of the guidance law used and there was always the question of - is it the best we can do with this LV configuration. This paper suggests to introduce a trajectory optimization program to calculate the optimal trajectory [3] and performance of a given LV configuration as part of the LV multi-disciplinary design process. The fast optimization program proposed in this paper allows to calculate the performance and sensitivity of a given LV configuration. Getting the estimated performance of the current configuration can stop the iteration process if the results satisfactory, or what's the direction to step if improvement is need according to the sensitivity analysis. The program proposed solves the optimal control problem of a LV in 3 main scenarios - 1) an expandable first stage, 2) booster lands on a drone ship down range of the launch site, 3) booster return to launch site. The option to reuse the booster, the most expensive component of a LV, is the most prominent technique to reduce launch cost that was proved to be feasible. SpaceX, using their Falcon 9 LV, has shown that by reusing the booster they can drastically lower the cost of launching to space. This can be particularly seen in the main parameter widely considered to describe the cost of launching - the price of launching 1 kg of mass into orbit by a launch system. Figure 1 shows this parameter on various LVs and it can be seen that Falcon 9 and Falcon heavy is in a league of their own. Many other companies are trying to design a LV with similar concepts. SpaceX is pushing the re-usability goal one step further by trying to reuse the second stage in addition to the first stage in the Starship launch system. Previous work that was done in this subject include the work by Ma et al. [4] - they present an optimization of a vertical landing reusable rockets, by using simple physical models and show the kinematics of such trajectories. Dresia et al.[5] use genetic algorithm based MDO to compare propellants types for partially reusable two-stage LVs. Designing a tool that is a one stop for all the performance and sensitivity analysis of a LV under different booster recovery strategies can make the iterative design process of a recoverable LV much faster.

3 Problem Formulation

Solving the Optimal Control Problem (OCP) of launching a satellite into orbit can be extremely challenging. In order for this problem to be used as a design tool for LV preliminary design and sizing, we first need to understand the main parameters that determine a LV performance. There are 2 main parameters that determine the overall performance of a LV - 1) the total useful mass ¹ that a LV can put into Low Earth Orbit (LEO) and 2) the total useful mass a LV can put into a high energy orbits (like Trans Lunar Injection, TLI). There is a can be a big difference between these parameters for different LV configurations. In this paper we will focus on a 2 stage LV which is the most popular design concept use in modern LVs. In order to focus on the most important parts of the space launch trajectory design, we shall make the following assumption:

¹The mass of the satellite, meaning the total mass reached the desired orbit minus the inert mass of the LV that is disposed

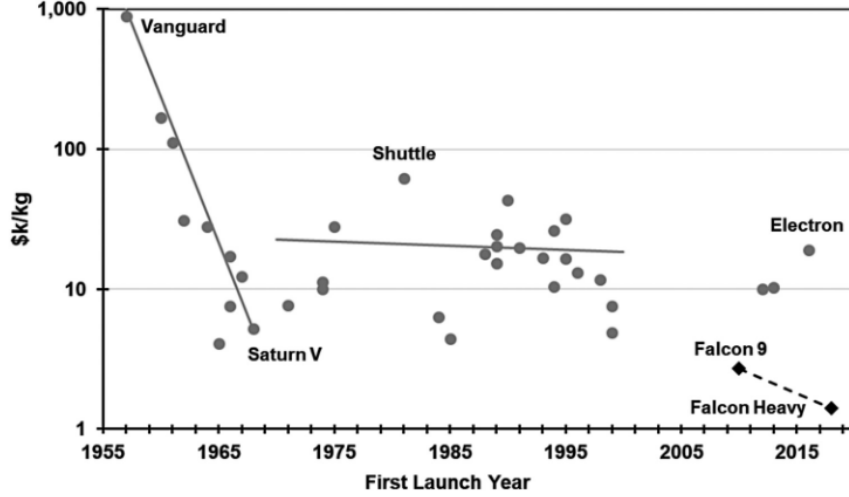


Figure 1: Launch Cost for different LVs, \$ per 1Kg. Taken from [6]

- The earth is a round sphere
- Standard Atmosphere
- No Winds

These basic assumption makes the mathematical description of the physical models easier while maintaining the dominant physical concepts that describe the problem. The booster recovery strategies examined in this paper include the following:

- **Expendable:** The booster is not recovered and may expend its full propellant load to maximize second-stage performance prior to separation.
- **ASDS:** Autonomous Spaceport Drone Ship recovery. The booster lands downrange on a GPS-station-kept barge. This imposes a modest performance penalty but higher logistical complexity; the ship's position is planned prelaunch to minimize booster divert effort.
- **RTLS:** Return to Launch Site. The booster flies back to the launch site, incurring a larger performance penalty but minimal logistics.

The LV two stage phase goes through very different regimes throughout the flight. At first, the boost phase is a powered ascent phase through the atmosphere that is relatively short and produces high aerodynamic loads. The second stage is relatively long and mostly flown in the vacuum of space where the mechanical loads are minimal. At each stage, a different equation of motion will be used to best describe the dynamics of the problem. The booster return phase starts at the vacuum of space and ends in a high structural and thermal load atmospheric reentry. Modeling the main parameters that affect these loads is a crucial part of the problem description and optimization.

4 Physical Models

The engineering complexity of a launch vehicle is very high due to 2 main factors. The first is the large numbers of engineering disciplines that one has to master in order to design a LV. The second one is the fact that in order to maximize the performance of a LV the safety factors that are used are minimized, which leaves no margins for errors. These two factors make the design of a LV extremely challenging. The kinematics of a space mechanics determine that in order to achieve orbital velocity, a satellite has to move very fast - above 8[km/s]. Maintaining these extreme velocity can be only done outside earth atmosphere where aerodynamic drag and heating associated with high speeds are not an issue. This means that a LV first has to reach high altitude as fast as possible in order to pay the minimal cost associated with heating and mechanical stresses in the LV. There are 3 main force acting on a LV during the launch process. First is the thrust produced by the rocket engine. Secondly is the gravitational force acting on the LV. Third is the aerodynamic forces acting on the LV - which are dominant mainly during the first phases of the flight.

4.1 Coordinate Systems

This paper will use two main Coordinate Systems (CS). The first will be used during the boost phase where most of the motion is done in a 2D plane. The LV lifts off from the launch pad and starts a pitch-over maneuver at some azimuth. The CS used in this phase is a 2D frame in which the origin is at the launch site and it attached to earth surface. the X direction points to the azimuth of flight and Z direction points up at the launch site. This CS is called Local Level Azimuth-Aligned frame (LLAA). See figure 2 for illustration. During the the second stage, the LV has to converge to the desired orbit which can produce maneuvers at all direction. For this reason, the CS used is Earth Centered Inertial (ECI). This is an inertial CS in which the origin is at earth center. X axis is at the equatorial plane and point to Greenwich at the time of lift-off, Z axis point to the north pole and Y axis completes a right hand CS. A procedure to transform the position a velocity vector from one CS to the other is presented in Appendix ??.

4.2 Rocket Engine - Thrust

In order to reach orbital velocities LV uses chemical rocket engines. There are many types of rocket engines. The most popular for LVs are solid and liquid rocket engines with each type having its own pros and cons. Liquid rocket engines are more efficient but weigh more and produce less thrust relative to solid rocket engines. In addition, liquid rocket engines are controllable and can be reignite on demand relative to solid rocket engines which have constant thrust profile and are One Shot Device (OSD). Choosing the rocket engine type is a major design decision of a LV. Most modern LVs use liquid rocket engines (some use solid rocket engines during the initial boost phase to increase the initial thrust at liftoff). This paper focuses on a two stage LV with both stages using a liquid rocket engines - the most popular design choice of modern LV (SpaceX Falcon 9, SpaceX Starship, Blue Origin New

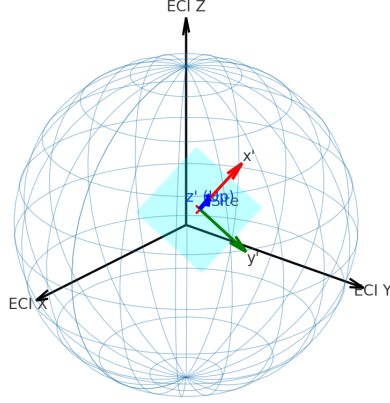


Figure 2: ECI coordinate system with local heading-aligned frame at a launch site. The tangent plane ($x'y'$) is shown in cyan , and z' (Up) is blue.

Glenn, Rocket Lab Neutron and others). To model the thrust of a rocket engine, the well known rocket engine equation ([7], chapter 3) is used:

$$T = \dot{m}V_e + (P_e - P_a)A_E \quad (1)$$

where, $\dot{m}$ is the mass flow rate through the engine, V_e is the exit velocity of the propellant from the engine nozzle, P_e is the propellant static pressure at the nozzle exit, P_a is the ambient pressure which is altitude dependent and A_e is the nozzle exit. For a given rocket engine geometry and if flow separation is not present (see Summerfield criterion [8]), the following connections can be made from isentropic flow theory for an ideal gas:

$$V_e = \sqrt{\frac{2\gamma}{\gamma-1}RT_c \left[1 - \left(\frac{P_e}{P_c} \right)^{\frac{\gamma-1}{\gamma}} \right]} \quad (2a)$$

$$P_e = P_c \left[1 + \frac{\gamma-1}{\gamma}M_e^2 \right]^{\frac{\gamma+1}{2(\gamma-1)}} \quad (2b)$$

Where T_c is chamber temperature, γ and R are the specific heat ratio and specific gas constant of exhaust gases, respectively. M_e is the exit Mach number of the exhaust which depends on nozzle geometry - expansion ratio $\epsilon = \frac{A_e}{A_t}$. All these parameters are constant

during engine operation, which means that V_e and $\frac{P_e}{P_c}$ stays constant during normal engine operation (no flow separation). The engine chamber pressure P_c is linearly connected to the mass flow:

$$\dot{m} = \frac{P_c A_t}{c^*} \quad (3)$$

where, A_t is the throat area of the engine nozzle and c^* is the property of the propellant. This means that P_e is linear with mass flow $\dot{m}$. To summarize the insights from equation 1-3, we can say that the thrust of a given rocket engine is dependent on 2 main parameters - 1) the mass flow through the engine (can be controlled by throttling) which has 2 effects - it directly changing the thrust by changing $\dot{m}$ in equation 1 and indirectly changing thrust due to chamber pressure changes which effect the exhaust static exit pressure P_e . 2) The ambient pressure which depends on altitude. To get the final model of the thrust a linear throttling model is assumed. The mass flow through the engine can be controlled. The throttling factor is defined as the ratio of the current mass flow relative to the maximal:

$$f_T = \frac{\dot{m}}{\dot{m}_{max}}, \quad f_T \in [f_T^{min}, 1] \quad (4)$$

where $\dot{m}_{max}$ is the maximal allowed mass flow through the engine and f_T^{min} is the minimal allowed throttling factor to maintain normal engine operation. Combining all into the thrust equation:

$$T(f_T, P_a) = f_T \cdot \dot{m}_{max} \cdot (V_e + \alpha) - P_a A_E \quad (5)$$

where the parameter α is a property of the engine:

$$\alpha = \frac{c^*}{A_t} \cdot \left[1 + \frac{\gamma - 1}{\gamma} M_e^2 \right]^{\frac{\gamma+1}{2(\gamma-1)}} \quad (6)$$

Usually, the engine is tested at sea level and vacuum, and therefore the data in these two points is known well and the performance is described by the maximal thrust and Specific Impulse (ISP):

$$Isp = \frac{T}{\dot{m} \cdot g_0} \quad (7)$$

Rearranging the thrust and mass flow equation to be described in these terms gives:

$$T(f_T, P_a) = f_T \left[T_{Vac} - (T_{Vac} - T_{SL}) \frac{P_a}{P_{SL}} \right] \quad (8a)$$

$$\dot{m} = f_T \cdot \frac{T_{Vac}}{ISP_{Vac} \cdot g_0} \quad (8b)$$

Equations (8) provide a practical means of predicting thrust and mass flow over the full operating envelope, using only a small set of test-derived parameters. In this formulation, T_{Vac} and T_{SL} encapsulate the combined effects of nozzle geometry, propellant thermochemistry,

and expansion losses at different ambient pressures, while $I_{sp, \text{vac}}$ links the propellant flow rate directly to measurable performance. Integrating equation (8) over the burn time of the engine give the famous rocket equation:

$$\Delta V = g_0 \cdot ISP \cdot \ln \frac{m_f + m_p}{m_f} \quad (9)$$

where here ISP is the average ISP along the engine burn, and m_p , m_f are the mass at burnout and propellant mass used during the burn of the LV's engine, respectively. The throttle factor f_T enters linearly, consistent with the assumption of proportional variation of chamber pressure and mass flow. This model can be validated by fitting T_{vac} , T_{SL} , and $I_{sp, \text{vac}}$ to static-fire data at sea level and altitude-simulation test conditions, and then comparing predicted thrust at intermediate f_T values and ambient pressures to measured flight data. Deviations from the model are expected if flow separation occurs at low ambient pressures, or if nonlinearities in injector or turbo-pump performance appear at extreme throttles.

4.3 Aerodynamic Model - Lift and Drag

During the boost phase of the launch vehicle (LV), aerodynamic forces act on its structure. The two primary aerodynamic forces are lift and drag. Lift is the component of the aerodynamic force acting perpendicular to the vehicle's velocity vector relative to the surrounding air, while drag is the component acting in the opposite direction to the velocity vector. A simple model for the lift force is a linear model:

$$F_L = \frac{1}{2} \rho V^2 \cdot S_{\text{ref}} \cdot C_{L\alpha} \cdot \alpha = Q_{\text{dyn}} \cdot S_{\text{ref}} \cdot C_{L\alpha} \cdot \alpha \quad (10)$$

where ρ is the density of the surrounding air, V is the velocity relative to air, S_{ref} is the reference area (usually taken as the LV cross-section area), $C_{L\alpha}$ is the lift line slope of the LV and α is the angle of attack - the angle between the velocity vector and the LV longitudinal body direction. In this paper α is used as a control input during the boost phase. This will allow to control the LV trajectory shaping during the boost phase. The linear lift model is a good approximation for low angles of attack. The drag force model consists from a shape drag at zero angle of attack and add induced drag:

$$F_D = Q_{\text{dyn}} \cdot S_{\text{ref}} \cdot (C_{D0} + C_{D_{\alpha^2}} \alpha^2) \quad (11)$$

where C_{D0} describing the shape drag of the vehicle and $C_{D_{\alpha^2}}$ is the induced drag coefficient. The drag coefficient is strongly dependent on the mach number. To take this into account, a piecewise quadratic fit for C_{D0} is used [9]:

$$C_{D0}(M) = \begin{cases} C_{D0} + k_1 M^2, & 0 \leq M \leq 0.8, \\ C_{Dt} + k_t(M - 0.8), & 0.8 < M \leq 1.2, \\ C_{Ds} + \frac{k_s}{M}, & 1.2 < M, \end{cases} \quad (12)$$

In addition, The aerodynamic model can provide an estimate to the aerodynamic loads associated with aerodynamic maneuver. When a aerodynamic body develops lift forces, it usually accompanied with significant structural bending moments which has to be limited (specially in a long body like LV). The bending moment for small angle of attack are proportional to:

$$M_{bending} \propto Q_{dyn} \cdot \alpha \quad (13)$$

4.4 Gravity Model

The gravity model used is the standard model of a uniform body. When expressed in ECI CS, the gravitational acceleration as a function of position vector $\vec{r}$ can be expressed as:

$$\vec{g}(r) = -\frac{\mu_e M_e}{r^3} \vec{r} = g_0 \cdot \frac{R_e^2}{r^3} \cdot \vec{r} \quad (14)$$

where μ is the gravitational constant, M_e and R_e is the mass and radius of earth, respectively and g_0 is the gravitational acceleration at earth surface. When expressed in local NED CS, the acceleration vector is as function of the position $[x, z]$ in NED is:

$$\begin{aligned} g_x &= -g_0 \frac{R_e^2}{r^2} \cdot \sin(\theta) \\ g_z &= g_0 \frac{R_e^2}{r^2} \cdot \cos(\theta) \end{aligned} \quad (15)$$

where $r = \sqrt{x^2 + (R_E - z)^2}$ and $\theta = \sin^{-1}(\frac{x}{R_e - z})$.

4.5 heating model

Aerodynamic heating is a primary driver of reentry loads, especially for a booster during the return phase. Boosters are typically oriented engine-first on entry because the nozzle/engine section is comparatively heat-tolerant; additional local thermal protection can be added to shield more sensitive structures. Even so, there is always a flight condition where the heat flux becomes excessive and risks structural damage, so operations enforce a heat-flux limit. In this work we model stagnation-point convective heating using the classical Sutton–Graves correlation [10]:

$$\dot{q}_c = \kappa_e \cdot \sqrt{\rho} v^3 \quad (16)$$

where, κ_e is an empirical coefficient that depends on booster structure properties, ρ is the air density and v is the velocity. The heat flux is has units of $\frac{Watt}{m^2}$. The allowed values for the maximal allowed heat flux can be approximated from theoretical calculations or computed from test flights - like telemetry from Falcon 9 flights. The max heating flux will constraint the dynamic condition during reentry. This can be managed by preforming engines burns to slow down the booster to limit the heat flux.

5 Equations Of Motions

The equation of motion that describe the launch of a typical launch vehicle can be divided into 2 phases. Since this paper talks about the recovery of the booster, an additional phase is added to model the reentry phase of the booster.

5.1 First Stage Ascent

The first stage start at stationary and upright position on the launch site (attached to earth surface). The trajectory of the LV during this phase is mainly planar - the LV pitch over to some azimuth (which is an optimization variable) and maintains this flight azimuth thought the phase. For this reason, the states of the LV during the first stage are:

$$X_1 = [x, z, v_x, v_z, m]^T \quad (17)$$

where, x, z are the horizontal and vertical position in the LLAA CS, respectively. v_x, v_z are the horizontal and vertical velocity in the LLAA CS respectively, and m is the mass of the LV. The inputs of the LV during this phase are:

$$\begin{aligned} U_1 &= [\alpha, f_T]^T \\ \alpha &\in [-\alpha_{min}, \alpha_{max}], \quad f_T \in [f_T^{min}, 1] \end{aligned} \quad (18)$$

where, α is the angle of attack of the LV, and f_T is the throttle of the engines. The dynamics of this phase is given as:

$$\begin{cases} \dot{x} = v_x \\ \dot{z} = v_z \\ \dot{v}_x = \frac{1}{m} [-\text{Drag} \cos(\gamma) - \text{Lift} \sin(\gamma) + \text{THRUST} \cos(\gamma + \alpha)] - g \sin(\theta_g) \\ \dot{v}_z = \frac{1}{m} [-\text{Drag} \sin(\gamma) - \text{Lift} \cos(\gamma) + \text{THRUST} \sin(\gamma + \alpha)] - g \cos(\theta_g) \\ \dot{m} = -\frac{\text{THR}_{\text{Vac}}^1}{\text{ISP}_{\text{Vac}}^1} f_T \end{cases} \quad (19)$$

where,

$$\begin{aligned} \gamma &= \text{atan2}(v_z, v_x) \\ \text{Drag} &= Q_{\text{dyn}} S_{\text{ref}} (C_{D_0} + C_{D_{\alpha^2}} \alpha^2) \\ \text{Lift} &= Q_{\text{dyn}} S_{\text{ref}} C_{L_{\alpha}} \alpha \\ g &= g_0 \frac{R_0^2}{x^2 + (R_0 + z)^2} \\ \theta_g &= \arcsin\left(\frac{x}{R_0 + z}\right) \end{aligned} \quad (20)$$

The dynamics above are valid only if the velocity of the LV is above some threshold (due to the definition of γ). For this reason, the initial condition of the optimization are defined after a section of an open loop vertical ascent for a few second. This logic is a popular operational technique that helps to clear the launch site infrastructure (launch tower).

5.2 Second Stage

The second stage phase begins after the physical separation between the first and second stage. This phase starts at high altitudes where dynamic pressure is negligible. During this phase the LV need to reach the orbital velocity (size and direction) that matches the desired orbit. this means that the second stage needs to maneuver in a 3D space. This phase is modeled in the ECI CS. The states of this phase are:

$$X_2 = [x, y, z, v_x, v_y, v_z, m]^T \quad (21)$$

where x, y, z are the position in ECI CS, v_x, v_y, v_z are the velocity in ECI and m is the mass. The inputs for the system during this phase are:

$$U_2 = [f_x, f_y, f_z]^T \quad (22)$$

where f_x, f_y, f_z are the forces produces by the engine in ECI frame. This means that the second stage can orientate itself in any angular state to point the engine's nozzle in the appropriate direction that produce the wanted inertial force. This is usually done by a actuators that control the second stage orientation (for example, gimbed nozzle). Since the stage engine can be throttled, the maximal force allowed is given by the maximal thrust that the rocket engine can provide:

$$\sqrt{f_x^2 + f_y^2 + f_z^2} \leq \text{THR}_{\text{vac}}^{(2)} \quad (23)$$

The dynamics of the phase is given by:

$$\begin{cases} \dot{x} = v_x \\ \dot{y} = v_y \\ \dot{z} = v_z \\ \dot{v}_x = \frac{1}{m} f_x - g_x \\ \dot{v}_y = \frac{1}{m} f_y - g_y \\ \dot{v}_z = \frac{1}{m} f_z - g_z \\ \dot{m} = -\frac{\sqrt{f_x^2 + f_y^2 + f_z^2}}{\text{ISP}_{\text{vac}}^2 g_0} \end{cases} \quad (24)$$

where,

$$\begin{aligned}
g_x &= g_0 \frac{R_0^2}{(x^2 + y^2 + z^2)^{\frac{3}{2}}} \cdot x \\
g_y &= g_0 \frac{R_0^2}{(x^2 + y^2 + z^2)^{\frac{3}{2}}} \cdot y \\
g_z &= g_0 \frac{R_0^2}{(x^2 + y^2 + z^2)^{\frac{3}{2}}} \cdot z
\end{aligned} \tag{25}$$

The second phase is initialized from the dynamic conditions of the first stage at the moment of separation. The function that transforms the state vector of the first phase to the state vector of the second phase is given in Appendix 10.

5.3 Booster Return

After the first stage completes its burn and separates from the second stage, the booster starts the recovery phase. There are two recovery profiles - performing a boostback burn after stage separation to land at the launching site or land on an autonomous platform down-range to the launch site. This entire phase is modeled at the same LLAA CS as the first stage. It is understood that this stage may be relatively long (a few minutes) and that the earth rotates underneath the booster during its flight. This is neglected since the goal of this simulation is an energetic performance evaluation of the LV. The accurate landing position of the rocket, or small maneuvers that the booster needs to do during the reentry phase are assumed to be energetically negligible and therefore ignored. The states of the phase are the same as the first stage, given in (17). The initial position and velocity of this stage are taken from the end of the first stage. The initial mass of the phase is given by the final mass of the first stage minus the mass of the second stage - the inert mass of the booster and any propellant mass that was left. If the booster needs to perform a boostback burn, which occurs at high altitude with negligible dynamic pressure, the inputs to the system are:

$$U_{boostback} = [f_x, f_z]^T \tag{26}$$

where, f_x, f_z are the forces in the x and z direction of the LLAA CS. This allows the booster to position itself at any direction to perform the boostback burn. Furthermore, since the booster energy management toward the landing is governed mainly by the engine thrust and aerodynamic drag, the aerodynamic lift forces are neglected during the reentry - it is assumed that the booster can't maneuver horizontally and is always at zero angle-of-attack. This means that the only input during the booster atmospheric reentry is the throttle of the engine opposite to the direction of the velocity. All the values for the parameters in this section are presented in Appendix 11.

6 Flight Logic and Constraints

The flight logic is the set of rules and order of operations that the LV performs during the launch. The LV starts stationary at the launch site. The flight logic and constraints of the first stage are:

- The LV starts the flight flying upright for a few second at full power (5 seconds is assumed).
- The maximal dynamic pressure during ascent can't be greater than Q_{max}^1 - this is to limit the mechanical and structural loads on the LV
- The lateral load on the vehicle due to aerodynamic maneuver ($Q_{dyn} \cdot \alpha$) could not exceed the allowed value - $Q_{dyn} \cdot \alpha \leq Q_{dyn}^{load}$. Due to LV and payload structural loads limits.
- The maximal axial acceleration of the LV should be kept below the allowed acc_{max} - mostly due to structural payload limits.
- The first stage end and stage separation has to be at a dynamic pressure lower than Q_{dyn}^{sep} - to ensure clean stage separation.

The flight logic and constraints of the second stage is:

- The maximal axial acceleration of the LV should be kept below the allowed acc_{max} - mostly due to structural payload limits.
- Payload Fairing separation is allowed a dynamic pressure below - $Q_{dyn}^{fairing}$
- The perigee of the final orbit of the second stage must be greater than $P_{parking}$
- The inclination of the final orbit of the second stage must match to the target orbit
- The apogee of the final orbit of the second stage must match the perigee of the desired orbit.

The flight logic and constraints of the booster reentry is:

- The dynamic pressure during reentry must no exceed the allowed value $Q_{dyn}^{reentry}$ - due to mechanical loads on the booster
- The heat flux on the booster during reentry must not exceed the allowed value $\dot{q}_c^{reentry}$ - heating constraints on the booster
- the booster reach the earth surface at a velocity lower than allowed - v_{land} .

7 Simulation Results

This chapter will present the simulation results for a Falcon 9 and Starship based LVs for launching to a predefined orbit from Cape Canaveral Launch Complex (LC). The target orbits are summarized in table 1. Cape Canaveral LC is located at the following location:

$$\text{Latitude} = 28.6^0, \quad \text{Longitude} = -80.6^0, \quad \text{Altitude} = 0[m\text{ASL}] \quad (27)$$

Cape Canaveral is used in this paper as the representative launch site (LC), with the caveat that it is not ideal for certain missions — most notably Sun-synchronous orbits (SSO), which require high-inclination trajectories and may be constrained by range-safety limitations or require dogleg maneuvers that reduce payload. This paper neglects range-safety requirements and assumes that all launch azimuths are possible without the need of a dog-lag maneuver. We consider a set of economically and scientifically important orbits that range from relatively low energy to high energy orbits:

- Low Earth Orbit (LEO) - circular (LEO), the most accessible orbit that requires the least amount of energy.
- Sun-synchronous Orbit (SSO) - near-polar LEO (inclinations are about $97-99^\circ$ at 500–800 km) that leverages Earth’s oblateness (J_2 gravitational term) so the orbital plane precesses $1^\circ/\text{day}$, maintaining constant local solar time over passes—ideal for imaging and remote sensing.
- Medium Earth Orbit (MEO) - MEO altitudes, in general, spans 2,000–20,000 km. The flagship example is GPS at 20,200 km and 55° inclination (12-hour period, “semi-synchronous”). Launch vehicles usually inject into an elliptical MEO transfer (low perigee, apogee near the target altitude); the spacecraft then raises perigee and circularizes at apogee.
- Geostationary/Transfer Orbit (GEO/GTO) - GEO is a circular, equatorial orbit at 35,786 km where spacecraft match Earth’s rotation and appear fixed over longitude — prime real estate for communications, weather and other purposes. From Cape, the standard launches target a GTO insertion (e.g., $200 \times 35,786$ km at $27-28.5^\circ$): the satellite performs apogee burns to raise perigee and reduce inclination, exploiting low velocity at apogee to minimize plane-change cost.
- Trans-Lunar Injection (TLI) - TLI departs a low-Earth parking orbit (often 200–300 km, 28.5° from Cape) onto an escape-energy trajectory intersecting the Moon’s vicinity.

The specific orbits used in this paper are summarized in Table 1.

The objective of the optimization is to maximize payload mass to the target orbit. The simulation uses publicly available data for the Falcon 9 and Starship launch vehicles (LVs); parameter values for each configuration are summarized in Appendix II. For each LV, the maximum payload is evaluated under the three booster recovery strategies presented in

Orbit Name	Perigee/Apogee Alt, Inclination	Remarks
LEO	200[Km]/200[Km], 28.6 [deg]	ISS orbit
SSO	780[Km]/780[Km], 98.6 [deg]	Sun-synchronous
MEO	1,1193[Km]/20,200[Km], 55 [deg]	GPS 55° transfer orbit
GTO	200[Km]/35,786[Km], 28.6 [deg]	transfer to GEO
TLI	200[Km]/384,400[Km], 28.6 [deg]	Trans-Lunar injection

Table 1: Typical parameters for common Earth orbits launched from Cape Canaveral LC

section 3. In addition to the recovery modes, we report an *operational* payload that accounts for real-world uncertainties (e.g., engine performance scatter, mass tolerances, atmospheric dispersion). As a rule of thumb, operational performance is computed by reducing the nominal engine specific impulse (Isp) by 2%. The optimization problem is formulated using Casadi [11] symbolic language in Python environment and solved using IPOPT [12] numerical solver. In that note, it is worth mentioning that since the problem formulation is non-Linear and Non-Convex (NLP), the optimal solution is in principal a local minima and there is no guarantee that this is the global optimal solution. The implantation of the optimal problem is given in [13]. The optimization results are summarized in Table 2.

Orbit Name	Recovery Strategy	Falcon 9 Block 5		Starship Block II	
		Nominal	Operational	Nominal	Operational
LEO	Expendable	24259	22649	268925	250641
	ASDS	21136	19663	230825	213438
	RTLS	17209	16050	202125	181139
SSO	Expendable	17740	16408	190794	174600
	ASDS	15223	14015	163791	143396
	RTLS	12132	11040	138323	116282
MEO	Expendable	9143	8227	90065	78362
	ASDS	7540	6717	69002	58104
	RTLS	5553	4853	50032	40052
GTO	Expendable	9247	8326	91982	80449
	ASDS	7633	6810	71317	60585
	RTLS	5630	4930	54119	50983
TLI	Expendable	6776	5990	61013	50983
	ASDS	5448	4746	43757	34457
	RTLS	3798	3201	29479	21080

Table 2: Nominal and operational payload mass [Kg] by target orbit and recovery strategy

The results show clear, monotonic payload loss with increasing mission energy demands (LEO $\rightarrow$ SSO $\rightarrow$ MEO/GTO $\rightarrow$ TLI) and with more aggressive recovery, as expected. Across both vehicles, ASDS recovery preserves most capability (about 15-20% penalty vs expendable), while RTLS imposes a large hit (about 30-50%), especially for GTO/TLI. The

operational derate (-2% Isp) translates to a payload reduction of a few percents for low energy mission profile up to about 7% of the more demanding missions. This highlights the importance of sizing to operational, not nominal, performance. Practically, ASDS seems to be the best recovery trade for most missions; RTLS is best reserved for low-energy targets; and very heavy or high-energy payloads may warrant expendable profiles or on-orbit refueling (for Starship) to recover margin.

The optimal launch of a Falcon 9 trajectory for MEO is presented for example. The first stage Dynamics is given in Figure 3. It can be seen the different trajectory for every recovery strategy. The length of the phase changes as a function of how much of the propellant needs to be spared for the recovery phase. The EXP and ASDS trajectories has a much more aggressive bending relative to the RTLS since, the RTLS profile optimizes between the will to bend fast (reduce gravitational losses) and to alleviate the BoostBack divert of the booster after separation. All mission profiles require the reduce the thrust factor during the region of peak aerodynamic pressure due to the structural constraints. The RTLS profile clears this region faster due to faster rate of climb. The thrust factor drops toward the end of the phase due to payload load constraints and the dynamic pressure separation constraint.

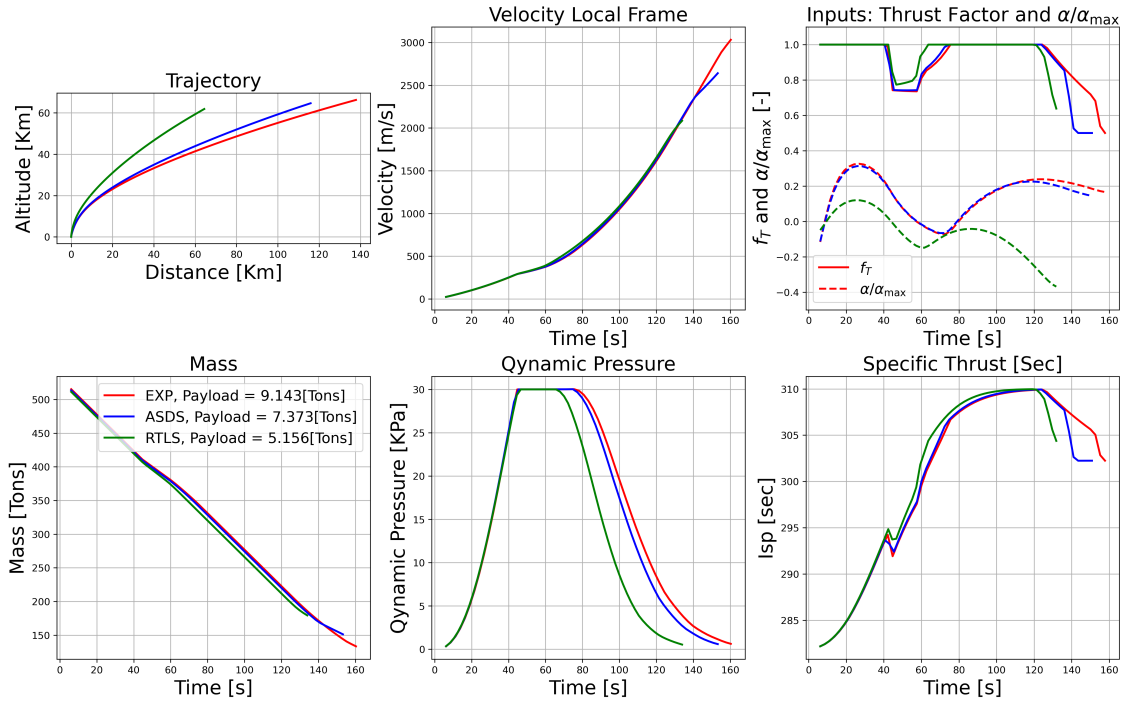


Figure 3: Flight dynamics of the first stage of a MEO target orbit for different booster recovery strategies

The second stage Dynamics is given in Figure 4. The initial conditions for the second stage is the final conditions for the first stage converted to ECI frame. It can be seen that the engine thrust factor is at maximum along the phase until the payload load constraint becomes active toward the end of the phase. In addition, the angle between the velocity

and thrust vector is presented (α) to show a rough measure of the trajectory shaping. In the mass profile graph the fairing separation can be seen when the conditions allow (altitude ≥ 100 [Km]). The orbital elements can be seen converging to the desired values of the parking orbit as defined in the flight logic.

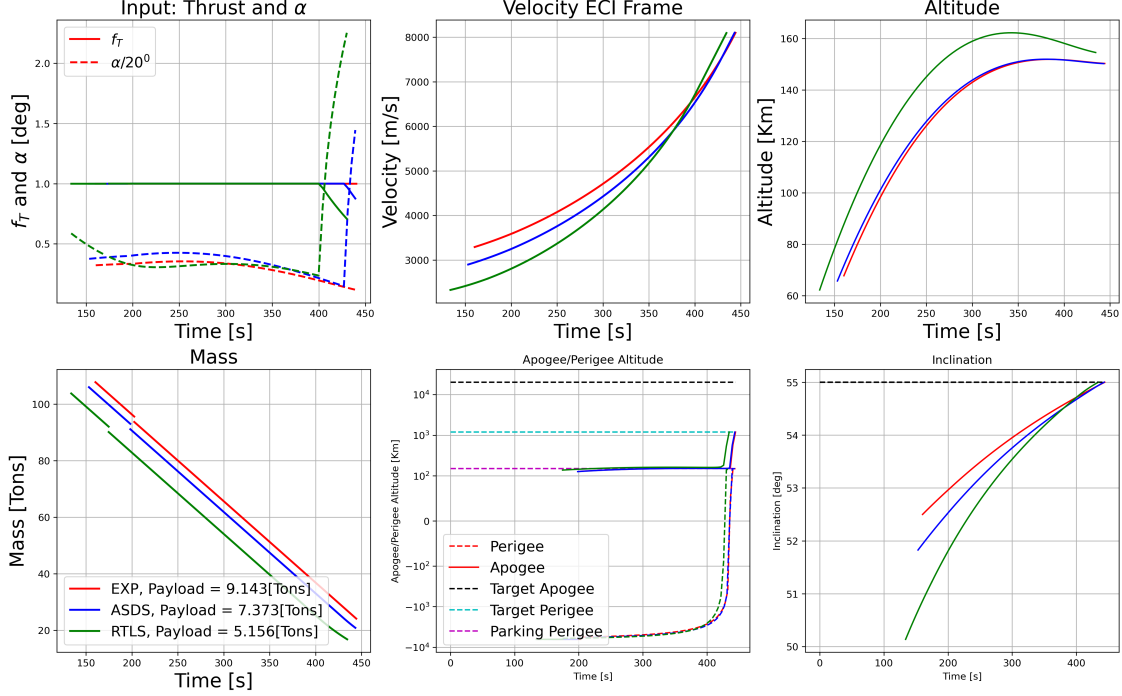


Figure 4: Flight dynamics of the second stage of a MEO target orbit for different booster recovery strategies

The return stage Dynamics is given in Figure 5. The initial conditions for the return stage is the final conditions for the first stage. It can be seen the engine operation whenever the booster mass is decreasing. In ASDS strategy there are 2 engines operations phases (reentry and landing burns), and in the RTLS strategy there are 3 engines operations phases (boostback, reentry and landing burns). It can be seen that the dynamic pressure and heat flux is limited to the desired values.

8 Sensitivity Analysis

This chapter analyzes the sensitivity of the LV performance to main design parameters of the LV. Each parameters has an engineering and economical importance when optimizing a LV design and it's very important to understand the sensitivity of every of these parameters. This analyses can also be used as an important engineering management tool. For example, if a company has a 1000 of engineering hours to spend on improving the LV performance, where they should invest it - improving the efficiency of the first stage engine performance

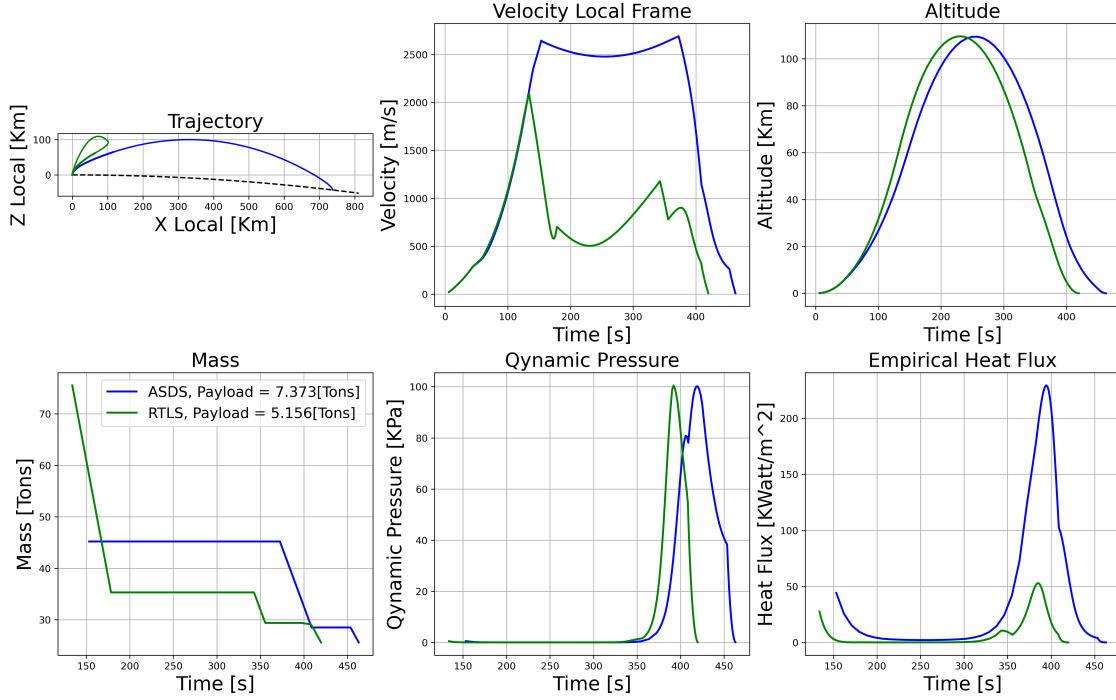


Figure 5: Flight dynamics of the Return phase of the booster for different recovery strategies

or reducing the inert mass of the second stage. The main parameters that are considered to be the most important to the overall performance of a LV are the following:

- **Engine Specific Impulse (Isp)** - The efficiency of the first and second stage rocket engine are one of the most critical factors influencing the performance of a LV. The engine ISP can be influenced by changing engine parameters like chamber pressure or increasing nozzle exit diameter.
- **Inert mass** - lowering the mass of the LV structure improves performance. The second stage inert mass is the most obvious and direct results - every Kg removes from the second stage inert mass, will increase payload capacity by a Kg
- **Engine Maximal Thrust** - most of the time, rocket engines operate at maximal capacity - the most efficient working point of an engine. Another important effect is the reduction of the gravitational losses during the powered stage. The faster a LV can get burn its propellant, the lower are the losses due to gravity pulling the LV on the other direction
- **Drag Coefficient** - C_{x_0} - The drag coefficient is the is a design parameter that reduces the performance of the first stage during decent
- **Stage Sizing** - The sizing of the stages is an important design parameter to optimize performance, especially in a recoverable booster configuration. choosing the right proportions between the stages is critical.

The analysis is done for 3 types of target orbits - LEO, MEO and TLI. The analysis shows the payload capacity of each of the LVs as a function of relative change in one of the design parameters relative to the nominal value presented in Appendix II.

8.1 Engine Specific Impulse (Isp)

Figure 6 shows the payload capacity as a function of relative change in Engine Specific Impulse (Isp). Stage 1 and stage 2 engines are analyzed separately. As expected, the payload capacity increase as the ISP increase. The increase is linear with Isp change for both stages. This is expected since the ΔV of a stage is linear with ISP for a specific stage - see (9). The sensitivity of the second stage to Isp change is higher than the first stage - this is why it is important to optimize upper engine performance. Starship sensitivity to Isp is greater than Falcon 9 due to different design point - essentially, the LV is too big for the Block 2 Raptor engines - This is most obvious in the TLI mission. The relative sensitivity between first and second stage is very different between Starship and Falcon 9 due to the different nominal operating point. The end velocity of the first stage of Falcon 9 is much higher than Starship. It can be seen that Falcon 9 has a ratio of 1.1 – 2 in sensitivity between the first stage and second stage. Where Starship has a ratio of 1.7 – 3. This is due to the fact that the Starship second is responsible for a much bigger portion of the final total velocity relative to the second stage of the falcon 9.

It can be seen that approximately the change in payload capacity is linear with the change in ISP. The following table summarizes the sensitivity of the payload capacity to the Isp at every scenario

Table 3: Payload mass sensitivity of Falcon 9 and Starship to ISP variations. Values show change in payload per +1% ISP increase for first and second stages.

Vehicle	Mission	Recovery	Nominal [Tons]	Δ Payload	Rel. Sens.	Δ Payload	Rel. Sens.
				+1% First-stage ISP [Tons]	(First) [%]	+1% Second-stage ISP [Tons]	(Second) [%]
Falcon 9	LEO	EXP	24.26	0.39	1.60	0.42	1.75
		ASDS	21.14	0.33	1.54	0.39	1.84
		RTLS	17.19	0.26	1.51	0.40	2.34
	MEO	EXP	9.14	0.18	1.99	0.28	3.10
		ASDS	7.54	0.15	2.04	0.27	3.52
		RTLS	5.55	0.11	2.06	0.24	4.39
	TLI	EXP	6.78	0.15	2.20	0.25	3.71
		ASDS	5.45	0.12	2.28	0.23	4.28
		RTLS	3.80	0.09	2.45	0.20	5.38
Starship	LEO	EXP	260.55	3.59	1.38	5.43	2.08
		ASDS	230.43	3.27	1.42	5.33	2.31
		RTLS	203.26	2.74	1.35	5.42	2.67
	MEO	EXP	85.50	1.86	2.17	3.91	4.57
		ASDS	68.81	1.67	2.42	3.76	5.47
		RTLS	52.58	1.36	2.59	3.63	6.90
	TLI	EXP	57.40	2.26	3.94	3.45	6.01
		ASDS	44.41	1.29	2.89	3.30	7.43
		RTLS	30.11	1.10	3.66	3.17	10.51

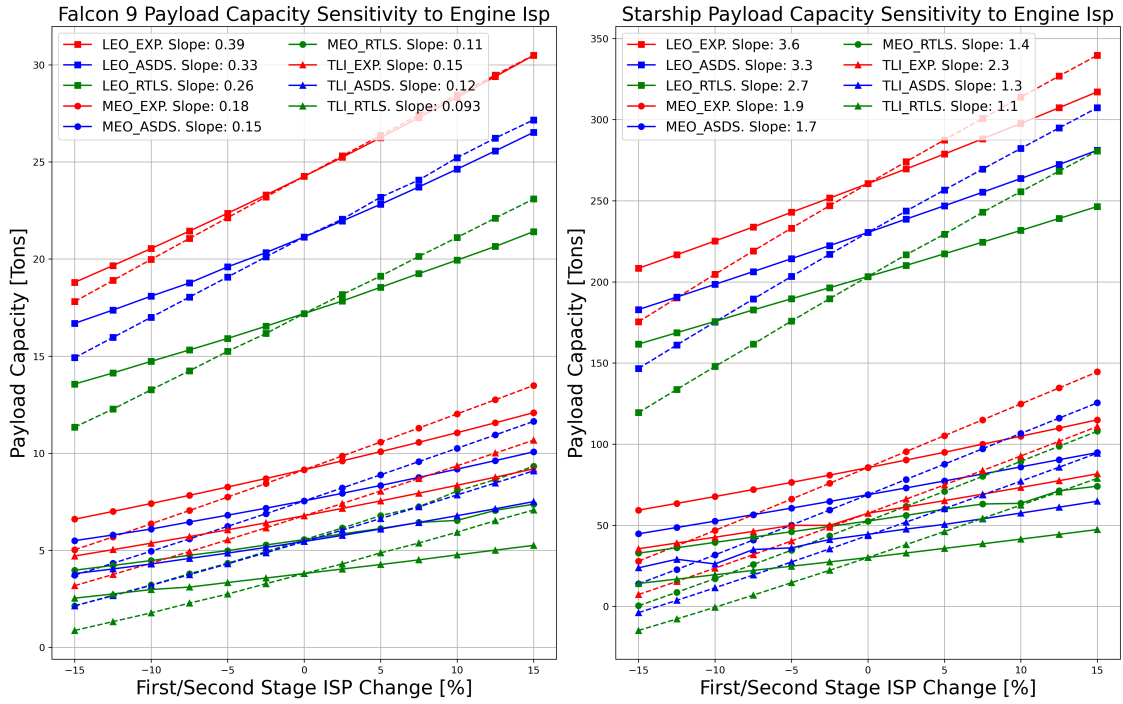


Figure 6: Payload capacity sensitivity of Falcon 9 and Starship to variations in engine Isp for both stages. Solid lines correspond to first stage Isp changes, dashed lines to second stage Isp changes.

8.2 Inert Mass

The inert mass of the LV is an important parameter of the design. Any mass that is not a propellant or payload is a mass that the engine need to push against and reduce performance. Most obvious is the second stage inert mass due to the fact that the second stage reach the final wanted orbit every inert 1 Kg that added to the second stage reduces the payload capacity by 1 Kg. The effort of reducing the inert mass of the LV is substantial in the design process. Figure 7 show the payload capacity for deviation from the nominal mass configuration. It can be seen that the payload capacity is much more sensitive to the second stage inert mass relative to the first stage.

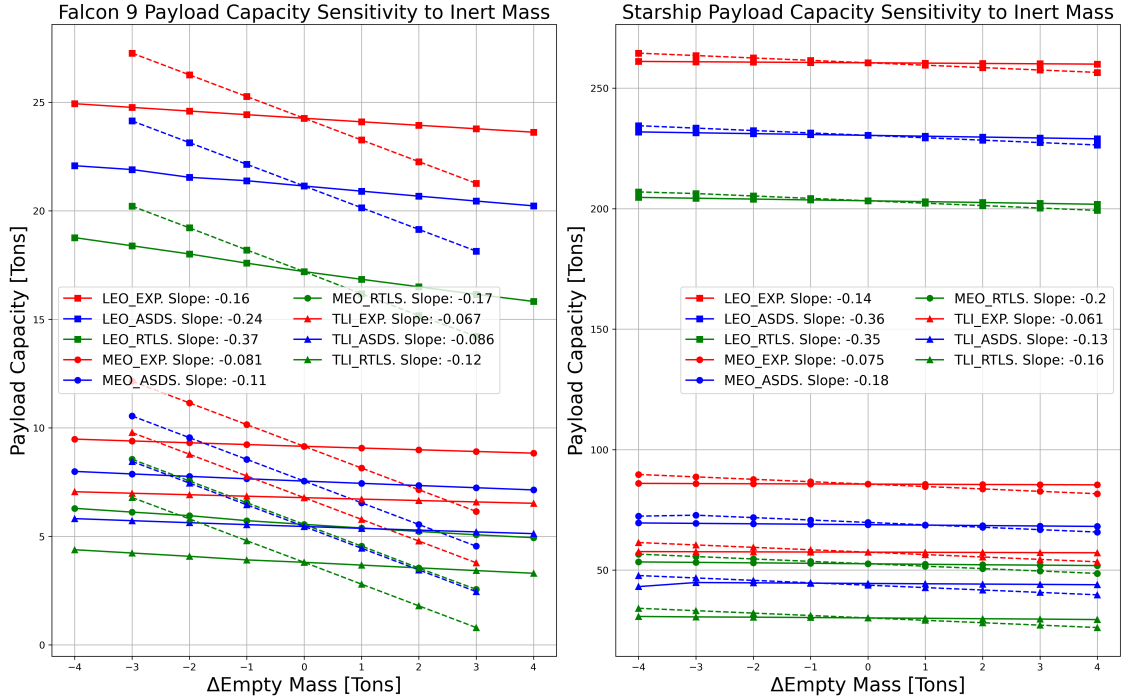


Figure 7: Payload capacity sensitivity of Falcon 9 and Starship to variations in Inert Mass for both stages. Solid lines correspond to first stage inert mass changes, dashed lines to second stage inert mass changes.

8.3 Engine Max Thrust

The maximal thrust is an important design parameter of the LV. The higher the thrust, the quicker the engines can accelerate the LV to the desired velocity. Or in other terms, higher thrust results in less engine burn time and thus reduces the loss due to gravity. Figure 8 shows the sensitivity of the payload capacity to the maximal thrust deviations relative to the nominal value. It can be seen that the sensitivity to the first stage thrust is much higher than the second stage. This is due to the fact that the gravity losses of the second stage

Table 4: Payload mass sensitivity to inert mass. Values show change in payload per +1 Ton increase.

Vehicle	Mission	Recovery	Nominal [Tons]	Δ Payload +1 Ton (First) [Tons]	Rel. Sens. (First) [%]	Δ Payload +1 Tons (Second) [Tons]	Rel. Sens. (Second) [%]
Falcon 9	LEO	EXP	24.26	-0.16	-0.68	-1.00	-4.12
		ASDS	21.14	-0.24	-1.14	-1.00	-4.73
		RTLS	17.19	-0.37	-2.16	-1.01	-5.88
	MEO	EXP	9.14	-0.08	-0.88	-1.00	-10.94
		ASDS	7.54	-0.11	-1.39	-1.00	-13.26
		RTLS	5.55	-0.17	-3.05	-1.00	-18.03
	TLI	EXP	6.78	-0.07	-0.98	-1.00	-14.76
		ASDS	5.45	-0.09	-1.57	-1.00	-18.36
		RTLS	3.80	-0.12	-3.14	-1.00	-26.33
Starship	LEO	EXP	260.55	-0.14	-0.05	-1.00	-0.38
		ASDS	230.43	-0.36	-0.15	-1.00	-0.43
		RTLS	203.26	-0.35	-0.17	-1.00	-0.49
	MEO	EXP	85.65	-0.08	-0.09	-1.00	-1.17
		ASDS	68.81	-0.18	-0.26	-1.00	-1.45
		RTLS	52.58	-0.20	-0.39	-1.00	-1.90
	TLI	EXP	57.40	-0.06	-0.11	-1.00	-1.74
		ASDS	44.41	-0.13	-0.29	-1.00	-2.25
		RTLS	30.11	-0.16	-0.53	-1.00	-3.32

are much higher - the LV starts from vertical climb (high gravity losses) and slowly starts to pitch over. Table 5 shows the sensitivity of the maximal thrust for both vehicles at each mission profile. The sensitivity of the Starship first stage thrust is very large - suggests that the examined Starship configuration is under-powered. The second stage of the falcon 9 shows very small sensitivity to the maximal thrust.

8.4 Propellant Mass

The propellant mass of each of the stages is a design parameter that influence the overall performance of the LV. Adding propellant mass to the first or second stage has a different effect on overall performance and trajectory shaping. Adding propellant to the first stage allows to accelerate the first stage to higher velocity but for every amount added, the LV needs to save some of it to be used during reentry due to the higher reentry. Adding propellant in the second stage makes the first stage push against heavier second stage. Figure 9 shows the payload capacity of both configurations to adding propellant to the first and second stage in different mission profiles. Table 6 shows the sensitivity of the payload capacity to added propellant mass at each stage for every mission profile.

8.5 First Stage C_{x_0}

The first stage drag of the LV determine the has an effect on the LV performance in two main parameters. The first is the direct losses that the LV exhibits during the first stage ascent.

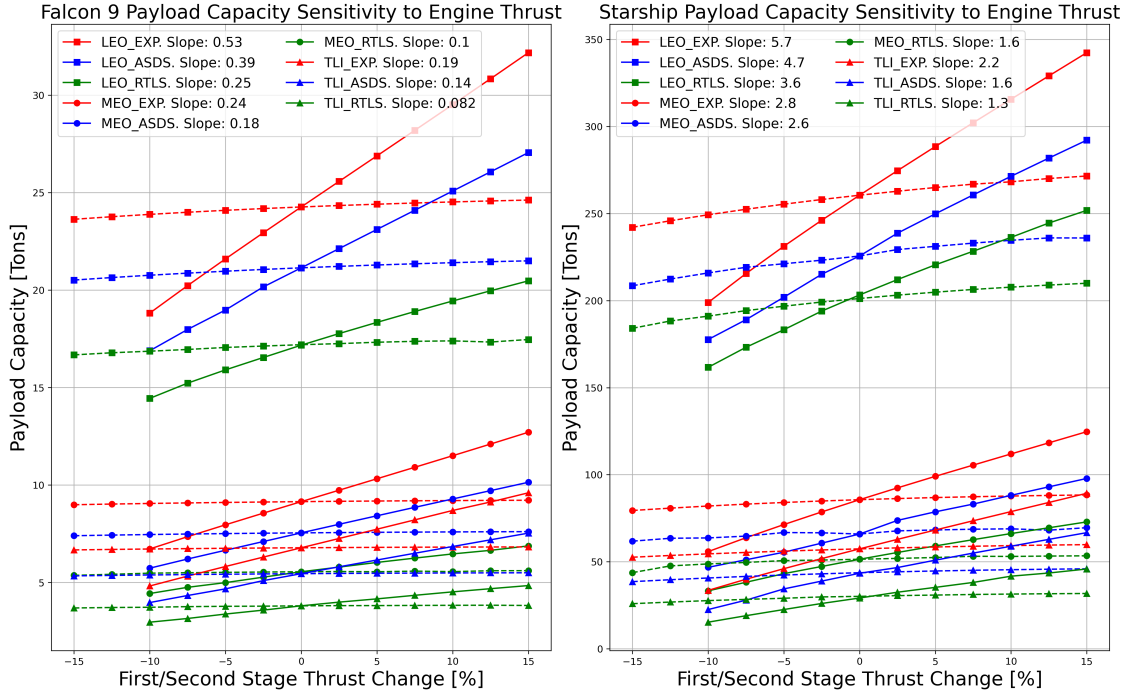


Figure 8: Payload capacity sensitivity of Falcon 9 and Starship to variations in Max Thrust for both stages. Solid lines correspond to first stage inert mass changes, dashed lines to second stage inert mass changes.

Table 5: Payload mass sensitivity to engine thrust variations. Values show change in payload per +1% thrust increase.

Vehicle	Mission	Recovery	Nominal [Tons]	Δ Payload +1% First-stage Thrust [Tons]	Rel. Sens. (First) [%]	Δ Payload +1% Second-stage Thrust [Tons]	Rel. Sens. (Second) [%]
Falcon 9	LEO	EXP	24.26	0.53	2.17	0.03	0.13
		ASDS	21.14	0.39	1.86	0.03	0.15
		RTLS	17.17	0.25	1.43	0.02	0.14
	MEO	EXP	9.14	0.24	2.57	0.01	0.08
		ASDS	7.54	0.18	2.35	0.01	0.09
		RTLS	5.55	0.10	1.86	0.01	0.10
	TLI	EXP	6.78	0.19	2.82	0.00	0.07
		ASDS	5.45	0.14	2.60	0.01	0.09
		RTLS	3.80	0.08	2.16	0.01	0.15
Starship	LEO	EXP	260.55	5.69	2.19	0.95	0.37
		ASDS	225.65	4.72	2.09	1.22	0.54
		RTLS	203.26	3.62	1.78	0.79	0.39
	MEO	EXP	85.65	2.76	3.22	0.28	0.33
		ASDS	65.92	2.61	3.96	0.22	0.34
		RTLS	51.43	1.61	3.12	0.19	0.37
	TLI	EXP	57.40	2.21	3.85	0.24	0.41
		ASDS	43.39	1.56	3.58	0.23	0.53
		RTLS	29.17	1.29	4.43	0.15	0.51

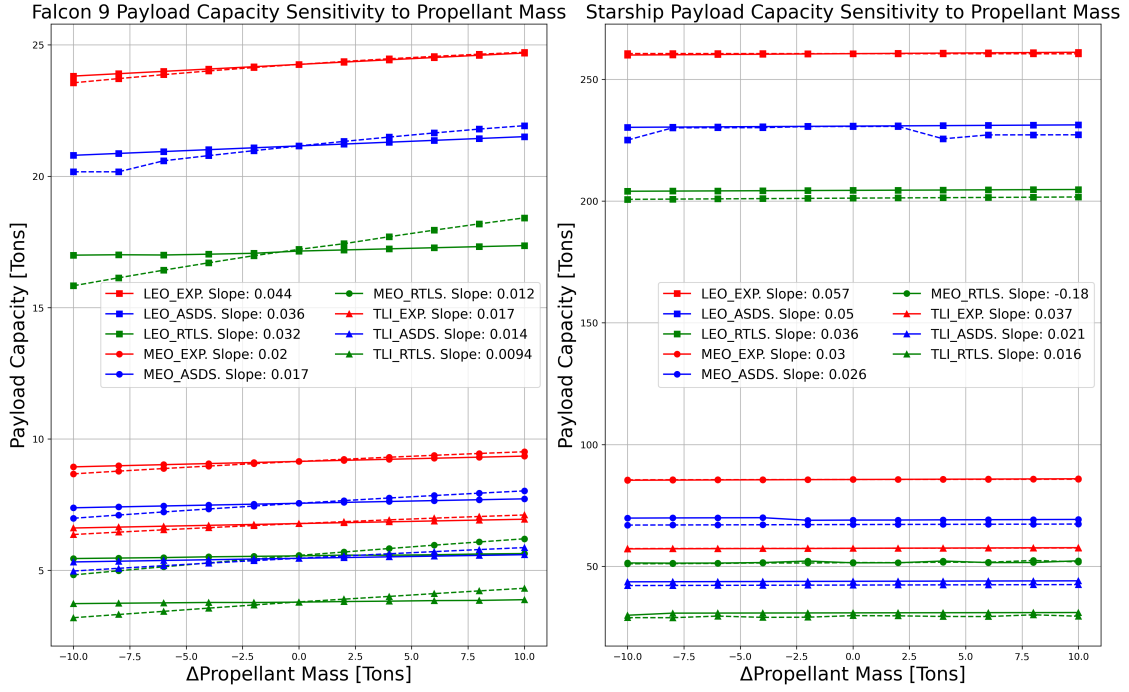


Figure 9: Payload capacity of Falcon 9 and Starship to variations in First/Second Stage Propellant Mass. Solid lines correspond to first stage inert mass changes, dashed lines to second stage inert mass changes.

Table 6: Payload mass sensitivity to propellant mass. Values show change in payload per +1 kg increase.

Vehicle	Mission	Recovery	Nominal [t]	Δ Payload +1 Ton (First) [t]	Rel. Sens. (First) [%]	Δ Payload +1 Ton (Second) [t]	Rel. Sens. (Second) [%]
Falcon 9	LEO	EXP	24.26	0.04	0.18	0.06	0.24
		ASDS	21.15	0.04	0.17	0.09	0.42
		RTLS	17.15	0.03	0.19	0.11	0.67
	MEO	EXP	9.14	0.02	0.22	0.04	0.46
		ASDS	7.55	0.02	0.23	0.05	0.69
		RTLS	5.55	0.01	0.21	0.07	1.21
	TLI	EXP	6.78	0.02	0.25	0.04	0.55
		ASDS	5.45	0.01	0.25	0.04	0.82
		RTLS	3.78	0.01	0.25	0.06	1.46
Starship	LEO	EXP	260.55	0.06	0.02	-0.01	-0.00
		ASDS	230.79	0.05	0.02	0.02	0.01
		RTLS	204.41	0.04	0.02	0.05	0.02
	MEO	EXP	85.65	0.03	0.04	0.01	0.01
		ASDS	68.99	0.03	0.04	0.02	0.03
		RTLS	51.43	-0.18	-0.34	0.04	0.08
	TLI	EXP	57.40	0.04	0.06	0.01	0.02
		ASDS	43.84	0.02	0.05	0.02	0.05
		RTLS	30.88	0.02	0.05	0.14	0.46

The second is the trajectory shaping of the first stage - the LV wants to pitch over as soon as possible to minimize the gravitational losses in one hand and climbing as fast as possible to reach low density altitude to minimize drag losses. The larger the drag coefficient, the slower can the LV pitch over. The drag of the LV is mainly determined by the shape of the LV fairing. Figure 10 shows the sensitivity of the payload capacity to the first stage drag for both vehicles.

8.6 Booster C_{x_0}

The booster drag coefficient determines how much propellant should be spared during the first stage ascent for the booster return phase. The larger the drag of the booster, the larger the aerodynamic deceleration of the booster during the reentry phase and less propellant is needed to slow down before landing. Figure 11 shows the payload capacity as a function of the booster drag relative to the nominal configuration. Table 8 shows the sensitivity of the payload to the booster drag for each mission profile and both LVs.

8.7 Launch Altitude

The launch altitude is usually not a design parameter - The launch site is given. Most launch sites are near the ocean due to safety considerations and usually there are no high mountains near the ocean. There is the possibility of launching from a flying platform (like

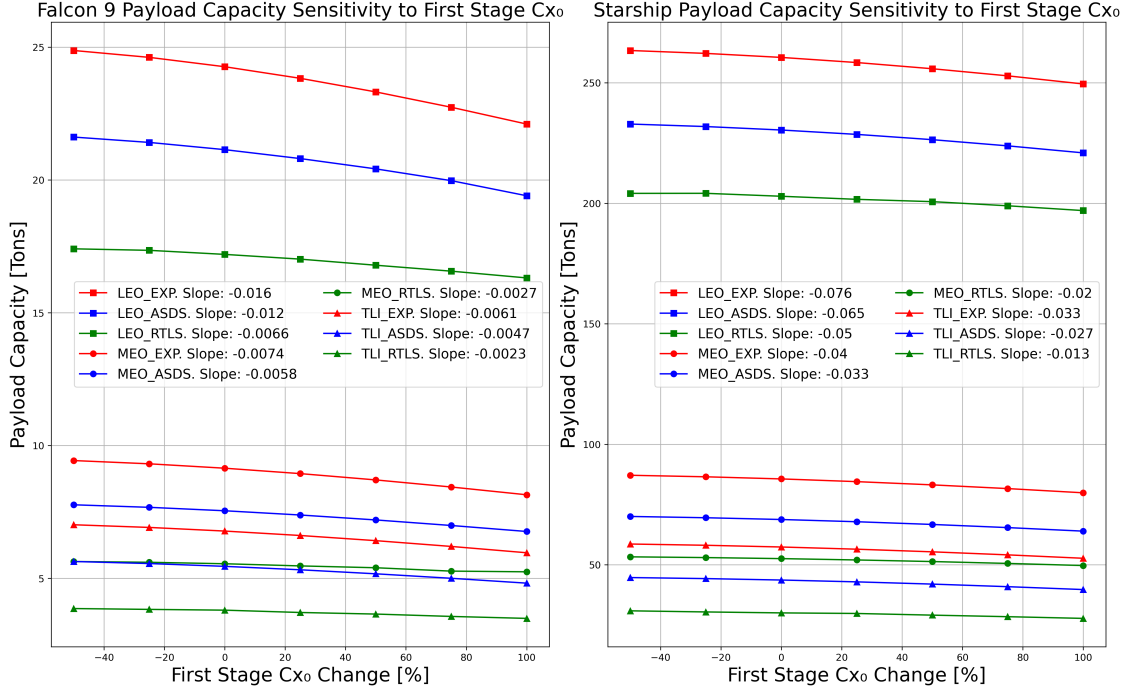


Figure 10: Payload capacity sensitivity of Falcon 9 and Starship to variations in First Stage Cx_0

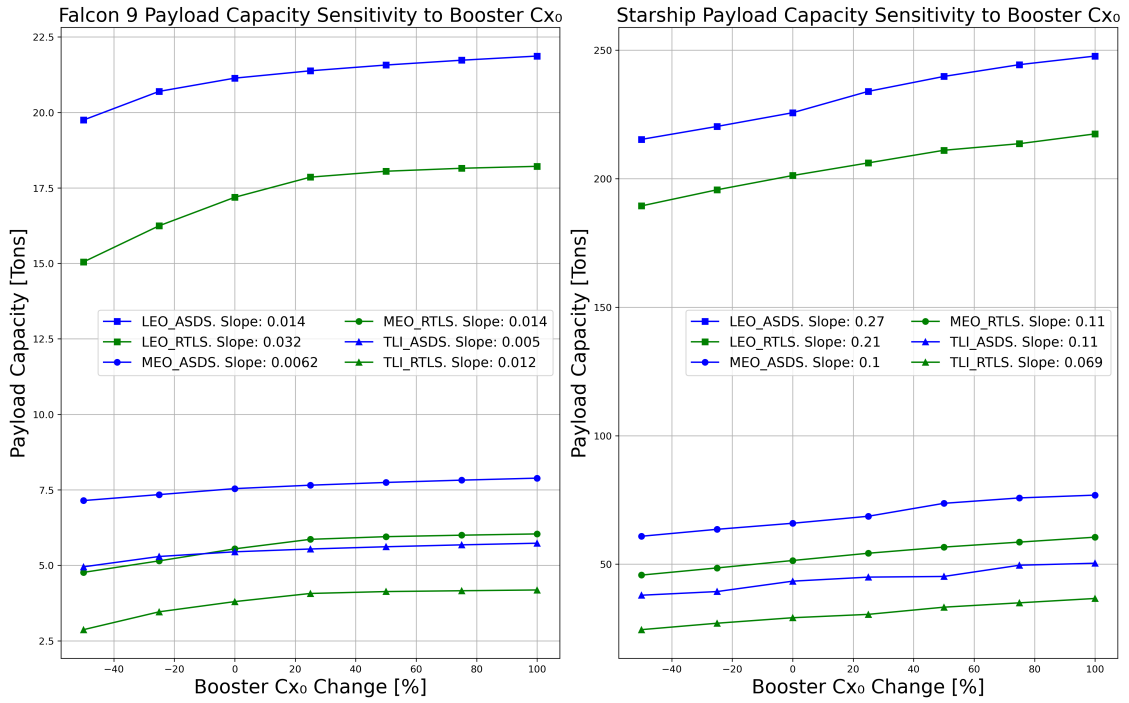


Figure 11: Payload capacity sensitivity of Falcon 9 and Starship to variations in First Stage Cx_0

Table 7: Payload mass sensitivity to baseline drag coefficient C_{x0} . Values show change in payload per +1% C_{x0} increase.

Vehicle	Mission	Recovery	Nominal [Tons]	Δ Payload +25% First-stage C_{x0} [Tons]	Rel. Sens. (First) [%]
Falcon 9	LEO	EXP	24.26	-0.16	-0.65
		ASDS	21.14	-0.12	-0.58
		RTLS	17.19	-0.07	-0.39
	MEO	EXP	9.14	-0.07	-0.81
		ASDS	7.54	-0.06	-0.77
		RTLS	5.55	-0.03	-0.49
	TLI	EXP	6.78	-0.06	-0.90
		ASDS	5.45	-0.05	-0.86
		RTLS	3.80	-0.02	-0.61
Starship	LEO	EXP	260.55	-0.76	-0.29
		ASDS	230.43	-0.65	-0.28
		RTLS	202.94	-0.50	-0.25
	MEO	EXP	85.65	-0.40	-0.47
		ASDS	68.81	-0.33	-0.48
		RTLS	52.58	-0.20	-0.38
	TLI	EXP	57.40	-0.33	-0.57
		ASDS	43.70	-0.27	-0.62
		RTLS	30.06	-0.13	-0.42

Table 8: Payload mass sensitivity to baseline Booster drag coefficient C_{x0} . Values show change in payload per +25% C_{x0} increase.

Vehicle	Mission	Recovery	Nominal [Tons]	Δ Payload +25% First-stage C_{x0} [Tons]	Rel. Sens. (First) [%]
Falcon 9	LEO	EXP	24.26	0.00	0.00
		ASDS	21.14	0.14	0.65
		RTLS	17.19	0.32	1.87
	MEO	EXP	9.14	0.00	0.00
		ASDS	7.54	0.06	0.83
		RTLS	5.55	0.14	2.57
	TLI	EXP	6.78	0.00	0.00
		ASDS	5.45	0.05	0.91
		RTLS	3.80	0.12	3.20
Starship	LEO	EXP	260.55	0.00	0.00
		ASDS	225.65	2.73	1.21
		RTLS	201.21	2.09	1.04
	MEO	EXP	85.65	0.00	0.00
		ASDS	65.92	1.01	1.54
		RTLS	51.43	1.14	2.22
	TLI	EXP	57.40	0.00	0.00
		ASDS	43.39	1.12	2.59
		RTLS	29.17	0.69	2.36

Stratolaunch) but these comes with considerable logistic problems and limits the weight of the LV. Nonetheless, it is important to understand the sensitivity. The advantages of launching as high as possible are trivial - less atmospheric layer to overcome during ascent and reduce gravitational losses. Figure ?? show the payload capacity when launching from different initial altitudes, and TABLE 9 shows the payload sensitivity to initial launch altitude.

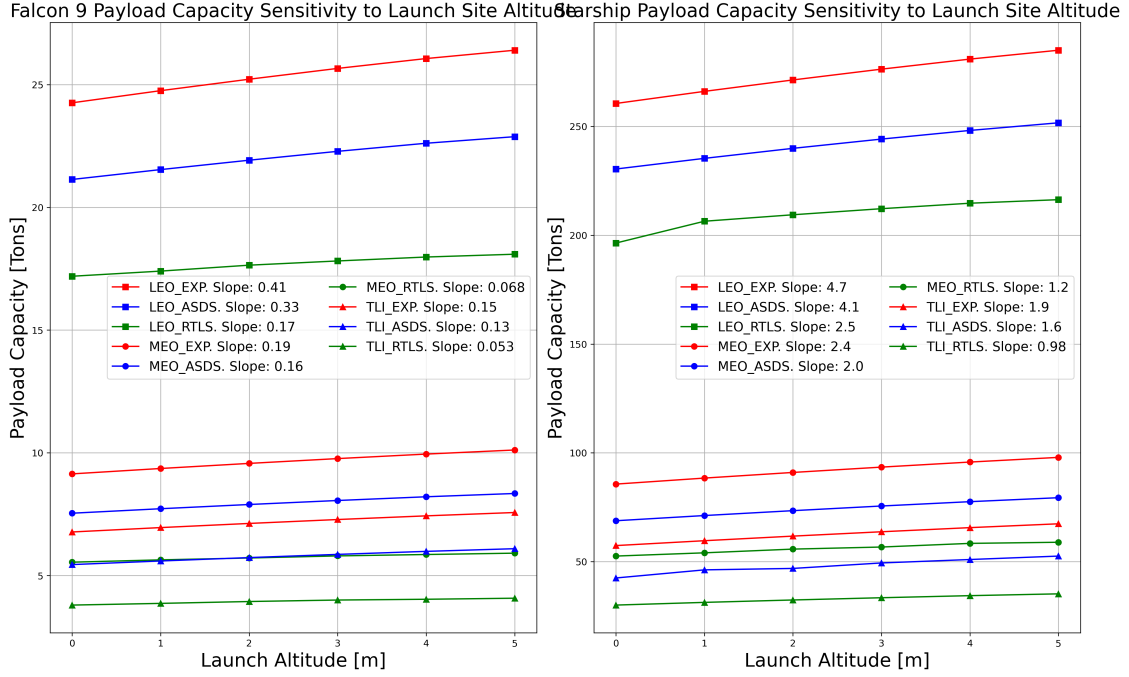


Figure 12: Payload capacity of Falcon 9 and Starship Vs Launch site altitude

8.8 Stage Partition

One of the main design parameters of a LV is how to partition the mass between the stages - how much propellant to put at every stage. There could be many considerations to take into account, other than pure performance - for exmaple, when making the first stage larger allows to recovery a greater portion of the LV. In this section we will only analyze the performance in terms of the payload capacity for a given total propellant mass. When moving propellant mass between stages, the inert mass also changes. The main inert mass of each stage is comprised from the structure of the propellant tanks. To take this into account, the mass fraction between the inert mass and the propellant mass from each stage is preserved relative to the nominal configuration:

$$\frac{m_I}{m_p}|_{stage_i} = constant, \quad i = 1, 2 \quad (28)$$

where m_I is the inert mass of the stage and m_p is the propellant mass of the stage. The Nominal propellant partition ration of the Falcon 9 is 81% (meaning, 81% of the total

Table 9: Payload mass sensitivity to launch-site altitude. Values show change in payload per +1000 m increase.

Vehicle	Mission	Recovery	Nominal [Tons]	Δ Payload +1Km LS Alt [Tons]	Rel. Sens. (First) [%]
Falcon 9	LEO	EXP	24.26	0.49	2.04
		ASDS	21.14	0.40	1.90
		RTLS	17.19	0.20	1.22
	MEO	EXP	9.14	0.22	2.40
		ASDS	7.54	0.18	2.42
		RTLS	5.55	0.09	1.70
	TLI	EXP	6.78	0.18	2.64
		ASDS	5.45	0.15	2.69
		RTLS	3.80	0.07	1.82
Starship	LEO	EXP	260.55	5.56	2.13
		ASDS	230.43	4.90	2.13
		RTLS	196.42	10.02	5.10
	MEO	EXP	85.65	2.76	3.23
		ASDS	68.81	2.39	3.47
		RTLS	52.58	1.49	2.83
	TLI	EXP	57.40	2.23	3.89
		ASDS	42.51	3.73	8.77
		RTLS	30.06	1.24	4.12

propellant mass is in the first stage). The Nominal propellant partition ratio of the Starship is 68.4%. Figure 13 shows the payload capacity as a function of the first stage propellant fraction - calculated $\pm 5\%$ around the nominal value of each configuration. It can be seen that Falcon 9 has its optimal propellant partition around the nominal value for the ASDS mission profiles - which is the most common mission profile for this LV, and probably the design point that the LV was optimized for. For the Starship configuration, the nominal partition is lower than the optimal partition for all mission profiles. The RTLS profile optimal partition is relatively close at 70% – 72%. The other mission profile has the optimal partition even higher. Its likely the the Starship design was optimized for other mission types that may include in-orbit refueling. Table 10 show the nominal design point payload capacity sensitivity to stage 1 propellant partition.

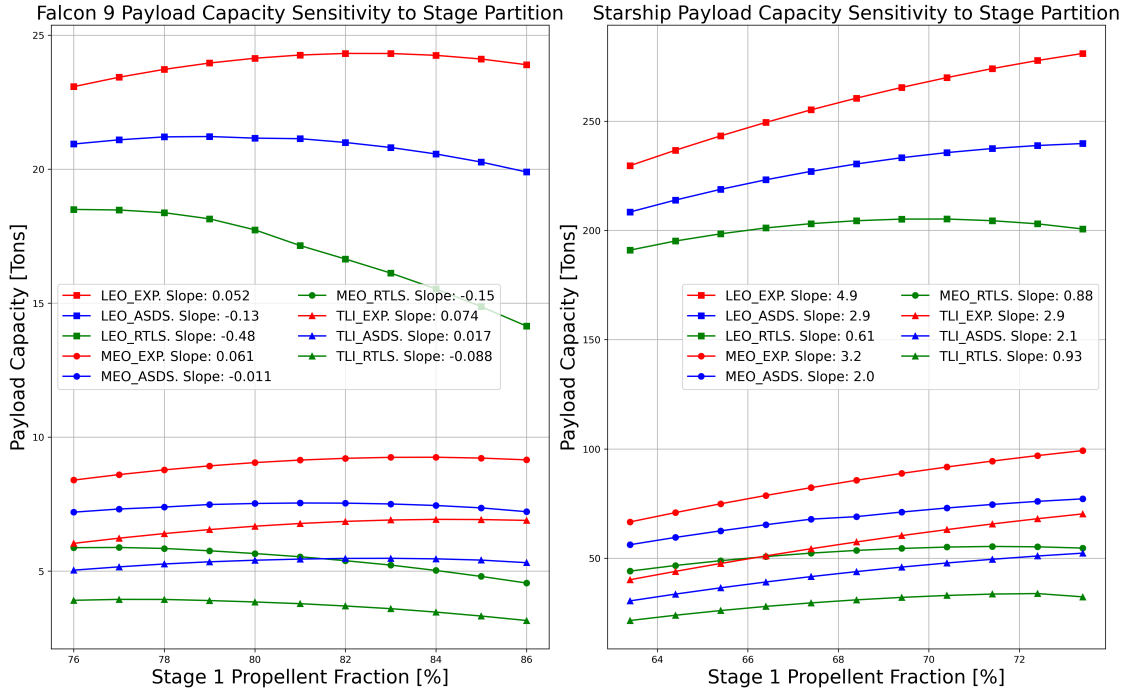


Figure 13: Payload capacity of Falcon 9 and Starship Vs Stage Partition

9 Conclusions

The paper presented an optimization framework that integrates the dominant physical models governing launch vehicle performance and constraints, and applies it to reusable booster configurations. By coupling trajectory optimization with sensitivity analysis, the method provides a one-stop preliminary design tool capable of capturing both performance and reusability trade-offs.

The results confirm that the Autonomous Spaceport Drone Ship (ASDS) recovery strategy provides the best compromise between payload performance and operational complexity,

Table 10: Payload mass sensitivity to first-stage propellant fraction. Values show change in payload per +1% increase.

Vehicle	Mission	Recovery	Nominal [t]	Δ Payload +1% First-stage Propellant Fraction [t]	Rel. Sens. (First) [%]
Falcon 9	LEO	EXP	24.26	0.01	0.04
		ASDS	21.14	-0.01	-0.04
		RTLS	17.15	-0.05	-0.32
	MEO	EXP	9.14	0.01	0.09
		ASDS	7.54	0.00	0.01
		RTLS	5.53	-0.01	-0.24
	TLI	EXP	6.78	0.01	0.13
		ASDS	5.45	0.00	0.06
		RTLS	3.78	-0.01	-0.20
Starship	LEO	EXP	260.55	0.51	0.20
		ASDS	230.43	0.31	0.14
		RTLS	204.45	0.10	0.05
	MEO	EXP	85.65	0.33	0.38
		ASDS	68.99	0.16	0.24
		RTLS	53.54	0.11	0.20
	TLI	EXP	57.40	0.30	0.52
		ASDS	43.82	0.22	0.50
		RTLS	30.91	0.13	0.41

typically incurring only a 15–20% penalty compared to expendable missions. Return-to-Launch-Site (RTLS) recovery imposes larger payload penalties (30–50%), making it best suited for low-energy targets or low mass payloads.

Sensitivity studies reveal important differences between Falcon 9 and Starship. Falcon 9 shows relatively low sensitivities across major design parameters, reflecting its maturity and balanced sizing. In contrast, the examined Starship Block II configuration exhibits strong sensitivities to first-stage thrust and stage partitioning, suggesting that further optimization or design refinement is required. Across both vehicles, second-stage performance and inert mass reductions were consistently shown to yield the highest impact on payload capacity.

Overall, this work demonstrates that trajectory optimization integrated with recovery modeling can serve as a powerful tool for guiding reusable launch vehicle design. Beyond quantifying payload trade-offs, it identifies where engineering investment is most valuable and provides a framework adaptable to new configurations, recovery architectures, and mission profiles.

10 Appendix I

The procedure to transform the position and velocity from the local CS used in the boost phase (LLAA) to the CS used in the second phase which is Earth Center Inertial (ECI) given launch azimuth is presented. The main differences between the coordinate systems are the position of the origin and the fact that ECI is inertial and LLAA is not. The main idea is to build a North-East-Up triad at the launch site, rotate the local state into ECI using the given azimuth, and then add the site’s ECI position and spin velocity—yielding the full inertial state to start the second-stage/ECI dynamics.

- Calculate the ECI position and velocity of the launch site based on the launch site Latitude (Lat_{LS}), Longitude (Lon_{LS}) and Altitude (Alt_{LS}):

$$\begin{aligned}\vec{P}_{LS} &= \begin{bmatrix} (R_e + Alt_{LS}) \cos(Lat_{LS}) \cos(Lon_{LS}) \\ (R_e + Alt_{LS}) \cos(Lat_{LS}) \sin(Lon_{LS}) \\ (R_e + Alt_{LS}) \sin(Lat_{LS}) \end{bmatrix} \\ \vec{V}_{LS} &= \omega_e \begin{bmatrix} -(R_e + Alt_{LS}) \cos(Lat_{LS}) \cos(Lon_{LS}) \\ (R_e + Alt_{LS}) \cos(Lat_{LS}) \sin(Lon_{LS}) \\ 0 \end{bmatrix}\end{aligned}\tag{29}$$

- build the North-East-Up vectors in ECI frame:

$$\begin{aligned}\hat{z}_{up} &= [\cos(Lat_{LS}) \cos(Lon_{LS}) \quad \cos(Lat_{LS}) \sin(Lon_{LS}) \quad \sin(Lat_{LS})]^T \\ \hat{x}_{north} &= [-\sin(Lat_{LS}) \cos(Lon_{LS}) \quad -\sin(Lat_{LS}) \sin(Lon_{LS}) \quad \cos(Lat_{LS})]^T \\ \hat{y}_{east} &= [-\sin(Lon_{LS}) \quad \cos(Lon_{LS}) \quad 0]^T\end{aligned}\tag{30}$$

- Rotate the north vector by angle Az to get the x vector pointed to the flight azimuth. y vector completes a right-hand CS with the Up vector :

$$\begin{aligned}\hat{x}_{Az} &= \cos(Az)\hat{x}_{north} + \sin(Az)\hat{y}_{east} \\ \hat{y}_{Az} &= \hat{z}_{up} \times \hat{x}_{Az} \hat{z}_{Az} = \hat{z}_{up}\end{aligned}\tag{31}$$

- The position and velocity in ECI frame as function of the position ($(\vec{P}_{LLAA})$) and velocity ($(\vec{V}_{LLAA})$) in LLAA frame :

$$\begin{aligned}\vec{P}_{ECI} &= P_{LS} + [\hat{x}_{Az}, \hat{y}_{Az}, \hat{z}_{Az}] \vec{P}_{LLAA} \\ \hat{y}_{Az} &= V_{LS} + [\hat{x}_{Az}, \hat{y}_{Az}, \hat{z}_{Az}] \vec{V}_{LLAA}\end{aligned}\tag{32}$$

11 Appendix II

This section presented the data used during the simulation and optimization of each of the LVs. The data was collection from open-source sources:

A Simulation Database

This appendix summarizes the database of parameters used in the launch vehicle simulations. Tables are divided by vehicle type: Falcon 9 and Starship. Each entry lists the parameter name, its symbolic representation, numerical value, physical units, and the corresponding source. The data for this paper was collected from Wikipedia [14], and other SpaceX official datasheet.

A.1 Falcon 9

This section presented the data used for the nominal configuration of Falcon 9 (Block 5) configuration.

Parameter Name	Symbol	Value	Units	Source
Diameter	D	3.66	m	SpaceX public data
Empty First Stage Mass	$m_{1,empty}$	25.6×10^3	kg	Simulation input
Empty Second Stage Mass	$m_{2,empty}$	4.0×10^3	kg	Simulation input
First Stage Propellant Mass	$m_{1,prop}$	395.7×10^3	kg	Simulation input
Second Stage Propellant Mass	$m_{2,prop}$	92.67×10^3	kg	Simulation input
Fairing Mass	m_{fair}	1.9×10^3	kg	SpaceX FCC filing
First Stage SL Isp	$I_{sp,SL}$	282	s	Merlin 1D specs
First Stage Vacuum Isp	$I_{sp,vac}$	310	s	Merlin 1D specs
First Stage SL Thrust	$T_{1,SL}$	$845 \times 10^3 \times 9$	N	Merlin 1D specs
First Stage Vacuum Thrust	$T_{1,vac}$	$914 \times 10^3 \times 9$	N	Merlin 1D specs
Second Stage Vacuum Isp	$I_{sp,2,vac}$	348	s	Merlin Vac specs

Second Stage Thrust	T_2	981×10^3	N	Merlin Vac specs
First Stage Max Q_{dyn}	$Q_{dyn}^{(1)}$	25×10^3	Pa	Approximation from telemetry
First Stage Max AOA	α_{max}	15	Deg	Assumption
Stage Separation Max Q_{dyn}	Q_{dyn}^{sep}	5×10^3	Pa	Approximation from telemetry
Payload Max Acceleration	a_{max}	4	g	Approximation from telemetry
Booster drag Coef.	C_{D0}^B	1.3	-	Approximation
Booster Max Dynamic Pressure	Q_{dyn}^B	100×10^3	Pa	Approximation from telemetry
Booster max heat flux	$\dot{q}_B$	100×10^3	$\frac{W}{m^2}$	Approximation from telemetry
Booster empirical heat flux coef.	κ	2×10^{-4}	-	Approximation
Earth Radius	R_E	6378137.0	m	WGS84
Earth Rotation Rate	ω_e	7.2921159×10^{-5}	rad/s	WGS84
Earth Grav. Parameter	μ	3.986×10^{14}	m^3/s^2	WGS84
Parking Orbit Perigee Alt.	h_{per}	150×10^3	m	Mission design

Table 11: Falcon 9 simulation parameters.

A.2 Starship

This section presented the data used for the nominal configuration of Starship (Block 2) configuration.

Parameter Name	Symbol	Value	Units	Source
Diameter	D	9.0	m	SpaceX public data
Empty First Stage Mass	$m_{1,empty}$	275×10^3	kg	Simulation input
Empty Second Stage Mass	$m_{2,empty}$	85×10^3	kg	Simulation input
First Stage Propellant Mass	$m_{1,prop}$	3250×10^3	kg	Simulation input
Second Stage Propellant Mass	$m_{2,prop}$	1500×10^3	kg	Simulation input
Fairing Mass	m_{fair}	0.0	kg	Not applicable
First Stage SL Isp	$I_{sp,SL}$	327	s	Raptor 2 specs
First Stage Vacuum Isp	$I_{sp,vac}$	347	s	Raptor 2 specs
First Stage SL Thrust	$T_{1,SL}$	$2256.3 \times 10^3 \times 33$	N	~33 Raptor 2 SL
First Stage Vacuum Thrust	$T_{1,vac}$	$2394.3 \times 10^3 \times 33$	N	Estimated
Second Stage Vacuum Isp (avg)	$I_{sp,2,vac}$	$\frac{3 \times 380 + 3 \times 347}{6}$	s	Raptor 2 SL+Vac m
Second Stage Thrust	T_2	$2300 \times 10^3 \times 3 + 2256.3 \times 10^3 \times 3$	N	3 Vac + 3 SL Raptor
First Stage Max Q_{dyn}	$Q_{dyn}^{(1)}$	25×10^3	Pa	Approximation
First Stage Max AOA	α_{max}	0.1745	rad	Constraint
Stage Separation Max Q_{dyn}	Q_{dyn}^{sep}	5×10^3	Pa	Approximation
Booster drag coef.	C_{D0}^B	1.1	-	With grid fins
Booster lift slope	$C'_{L,B}$	0.5	1/rad	Assumption
Booster Max Dynamic Pressure	Q_{dyn}^B	120×10^3	Pa	Approximation
Booster max heat flux	$\dot{q}_B$	150×10^3	$\frac{W}{m^2}$	Approximation
Booster empirical heat flux coef.	κ	2×10^{-4}	-	Model fit
Fairing Separation Altitude	h_{sep}	80×10^3	m	Simulation input

Earth Radius	R_E	6378137.0	m	WGS84
Earth Rotation Rate	ω_e	7.2921159×10^{-5}	rad/s	WGS84
Earth Grav. Parameter	μ	3.986×10^{14}	m^3/s^2	WGS84
Parking Orbit Perigee Alt.	h_{per}	150×10^3	m	Mission design

Table 12: Starship simulation parameters.

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